

Recap

Sniffers monitor network traffic via eavesdropping

Wireshark is a de-facto standard

Platforms to collect accurate Internet measurements

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

106

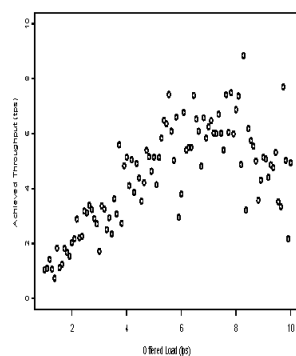
Measurements vs objectives

Measurements being collected must be consistent with the objectives of the monitoring project

Objective: to assess the impact on an existing infrastructure of the deployment of a video streaming service offered to users connected via a wireless network

Performance tests, i.e., measurements collected under an artificial load that reproduces the behavior of the users

Measurements show that the throughput decreases as the load increases (*baseline*)



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

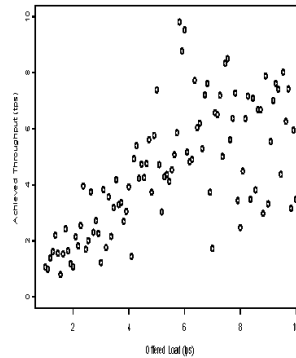
107

Measurements vs objectives (cont'd)

Conjecture: the wireless network is the bottleneck

The provider upgrades the infrastructure by adding two Access Points

Performance tests show that the throughput improves but not as much as it should



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

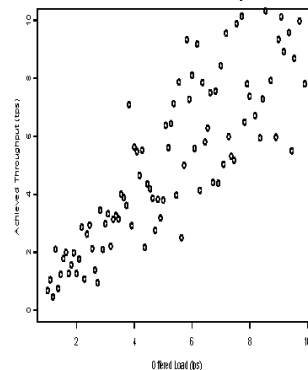
108

Measurements vs objectives

More detailed investigations (i.e., passive techniques combined with active techniques) that consider the entire infrastructure show that:

1. Packet loss rate is very small
2. RAM of the server is overutilized (i.e., bottleneck)

After the RAM upgrade, the throughput increases with the load as expected



© M. C. Calzarossa 2026

Enterprise Digital Infrastruct

109

Monitoring challenges

Monitoring overhead and intrusiveness might affect the operating conditions and performance of technological infrastructures and services

Both aggressive and conservative monitoring activities should be avoided

An appropriate trade-off between the amount of measurements to be collected and the “cost” associated with the measurement process has to be achieved

Unexpected effects might appear when monitoring activities are not performed in dedicated environments (e.g., speed tests, virtualized environments)

Experiments should be repeated multiple times to ensure the quality of the measurements

Legal, privacy and ethical issues (e.g., GDPR, “acceptable” usage policy) should also be considered very seriously

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

110

Measurement quality

Measurement quality is of paramount importance since measurements are often at the basis of many decision processes

It is necessary to assess the accuracy and precision of the selected monitoring tools (e.g., packet sniffer)

Changing insignificant and seemingly irrelevant aspects of the infrastructure used in the monitoring experiments can introduce a *bias* in the measurements (e.g., geographic coverage)

Representativeness and validity of the measurements should be ensured to avoid the risk of ignoring significant although *rare* phenomena

Sampling techniques should be applied with particular care (e.g., mice and elephant phenomenon)

Domain knowledge is a fundamental prerequisite for any monitoring activity

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

111